IPCC Tier 2 Livestock GHG Uncertainty Calculator

User Guide

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1 Overview

1.1 What this tool does

National GHG inventories based on IPCC Tier 2 methods require not only a central estimate of emissions but also a quantification of their uncertainty. This tool automates that process for livestock (cattle) using Monte Carlo simulation:

1. You supply your country-specific Tier 2 activity data and emission-factor inputs, each with an uncertainty range.
2. The tool randomly samples from the uncertainty distributions you have defined (thousands of times) and propagates those samples through the full IPCC Tier 2 equation chain (Eq. 10.1–10.34).
3. It reports the resulting 95% confidence interval around your emission estimate, identifies which parameters drive the most uncertainty (sensitivity analysis), and formats the results for submission in the format of IPCC 2006 Vol. 1 Ch. 3 Table 3.3.

The approach follows IPCC Approach 2 (Monte Carlo) as described in IPCC 2006 Vol. 1 Ch. 3.

1.2 Who this tool is for

This tool is designed for national inventory practitioners who:

- Are already conducting or have recently completed a Tier 2 cattle GHG inventory (enteric fermentation, manure management, direct and indirect N_2O from pasture deposition)
- Have at least rough estimates of the uncertainty in their key activity data and emission factors
- Need to report the uncertainty range for inclusion in national inventory reports

The tool assumes basic familiarity with IPCC Tier 2 methodology. Users do not need prior experience with Monte Carlo methods, but a conceptual understanding of probability distributions is helpful.

1.3 Emission sources covered

IPCC Category	Source
3.A.1	Enteric fermentation: cattle (CH_4)
3.B.1	Manure management: cattle (CH_4)
3.B.1	Manure management: cattle (N_2O direct)
3.B.5	Manure management: cattle (N_2O indirect)
3.D.1.c	Direct N_2O from pasture/range/paddock deposition
3.D.2	Indirect N_2O from pasture/range/paddock deposition

The Monte Carlo simulations can be run for enteric fermentation alone, or for enteric fermentation together with one or more (or all) of the other emission sources listed above. If you are interested to only run simulations for manure-related emission sources, you should supply country-specific Tier 2 input parameters for enteric fermentation. If these are not supplied, the tool will auto-fill

these inputs using IPCC default parameters and uncertainty ranges, which may not reflect the assumptions in your inventory.

2 Preparing your data

2.1 What you need before you start

Before opening the tool, assemble the following:

1. **Activity data**: For each animal sub-category, the population (N), body weight (BW), weight gain (WG), milk yield (for cows), milk fat %, digestible energy (DE%), manure management system (MMS) fractions, and other Tier 2 inputs.
2. **Uncertainty ranges**: For each parameter, an estimate of its uncertainty, typically either:
   - A percentage uncertainty (e.g. ±20% representing the half-width of a 95% confidence interval), or
   - Explicit lower and upper bounds.
3. **Manure management fractions** (if running manure management CH_4 or N_2O): The percentage of manure allocated to each management system (e.g. pasture, solid storage, liquid, etc.). If you are uncertain about the allocation itself, the tool uses a data uploading template in which the Manure_Management tab has optional `lower_fraction` / `upper_fraction` / `distribution_fraction` columns, fill them and the tool will sample each MMS within its range, renormalising every Monte Carlo iteration so the simplex (per-group rows sum to 100%) is preserved. The sampled values surface as `fraction_<mms>` in the sensitivity tornado so MMS allocation can show up as an influential driver. Leave blank to keep `fraction_pct` deterministic (the default, matching the IPCC Inventory Software).
4. **Emission factors and other coefficients**: If your inventory uses country-specific emission factors (e.g. EF3, EF4, etc.) or country-specific values of other parameters (e.g. C, Ca, Cp, Frac_GasMS, etc.), prepare these parameter values and the associated uncertainty estimates. If your inventory uses the IPCC default values, the tool can auto-fill these default values.

If you do not yet have uncertainty estimates, see Section 2.4 for guidance on obtaining them.

You only need the parameters used by the emission sources you select. The tool's pre-run check is source-aware: if you select only the methane sources (i.e. enteric fermentation, manure management CH_4), it will not ask you to fill the N_2O parameters, and vice versa. So a methane-only run never blocks on blank manure-N_2O cells. The managed-storage N_2O quantities (EF3 per system, Frac_GasMS, Frac_LeachMS) live in the Manure_Management tab, fill them there, not in the Parameters tab.

The tool uses a specific data upload template, described in the following sections. When you assemble your inputs to the tool, you may either input them directly into the template (which can be downloaded from the `1. Data input' tab), or you can use AI-assisted data preparation (see Section 2.2 below) to convert the data from your format into the template's format. In either case, it is useful to familiarise yourself with the parameter definitions and units used by the tool (see the`Definitions' tab) so that your inputs are

aligned with how the tool treats each parameter. For example, MMS fractions should be expressed as percentages (e.g. '5' meaning 5%) rather than fractions (e.g. 0.05).

2.2 AI-assisted data preparation

If your raw inventory data lives in your own Excel or CSV files with column names or other formatting that does not match the template, the tool offers an **in-app AI translator** that does the column-mapping and unit-conversion work for you. It is a small chat panel on the 'Resources' tab; you upload your data and the AI walks you through a three-step process that produces a filled template you can download.

The three steps (for each inventory). The translator works in three explicit passes: Explore, Clarify, Emit, instead of trying to do everything at once. This structure stops the AI from silently dropping rows or substituting catalogue defaults for values you supplied.

1. Open the **Resources tab**. The "AI Translator" card is the first item below the user manual, with a green left-edge accent and a short intro paragraph explaining what it does.
2. **Sign in with your email.** Enter your email address; you'll receive a one-time sign-in link by email. Every address is reviewed by the administrator before it is approved, including CGIAR addresses, because each translation is billed to the project (typical turnaround: same working day). Approval is needed once. The sign-in is then remembered indefinitely on this browser. To sign out, clear the site data in your browser settings.

   Non-CGIAR email: exact step-by-step:

   (a) Enter your email in the AI Translator card and click **Send sign-in link**.

   (b) Within 1–2 minutes, an email arrives in your inbox from `noreply@cattle-uncertainty.app` with subject *"Sign in to the IPCC Cattle GHG Uncertainty tool"*. Click the link inside (valid for 15 minutes).

   (c) The link opens the app and recognises that your email is not yet approved. You will see a green confirmation message: *"Thanks, your request to access the AI translator (your.email@example.org) has been forwarded to the administrator for approval. You'll receive an email once approved."* At the same moment, the administrator receives a notification email about your request.

   (d) The administrator reviews the request and adds your email to the approved-users list. Typical turnaround is the same working day.

   (e) Once approved, you receive a welcome email from the administrator confirming that you are now on the list. The email contains a direct link back to the app and a reminder of the workflow.

   (f) Return to the AI Translator card, enter the same email address again, and click **Send sign-in link**. A fresh magic-link email arrives within 1–2 minutes.

(g) Click the new link. This time the app confirms your access, drops a long-lived sign-in cookie in your browser, and the chat panel unlocks.

On subsequent visits from the same browser the cookie keeps you signed in: there is no need to repeat the magic-link step unless you have cleared site data.

3. **Step 1: Explore.** Upload your raw data file (`.xlsx` or `.csv`) using the file picker at the top of the chat panel. The AI reads every sheet and responds with a brief summary of what it found: the values it identified for each parameter and sub-category (with their units and bounds), and which parameters are missing and will fall back to IPCC defaults. Read the summary carefully to confirm it captured every parameter your file actually contains.

4. **Step 2: Clarify.** The AI then asks a short set of clarifying questions; answer them in plain English in the chat box (typically 2–5 short questions, e.g. *"Is your column* `weight` *body weight or mature weight?"*, or *"Your MMS row labelled 'PIT', does that map to* `liquid_slurry` *or* `solid_storage`*?"*). If the summary missed a parameter that IS in your file, point that out. If you want to add data the file doesn't have (population numbers, country-specific defaults), type them now. If you'd rather defer specific parameters to IPCC defaults, just say so. Note: when the AI asks a set of questions it may say to provide your answers and then click **Produce template now**; even so, it is a good idea to answer the questions first and confirm that the AI has understood your answers and has no further clarifications before you click it.

5. **Step 3: Emit.** When you have answered the AI's questions, click **"Produce template now"** (left-most button below the chat box). The AI mechanically translates the values it identified, the IPCC defaults for the missing parameters, and the biological zeros you confirmed into the template. A green **"Download template (.xlsx)"** button appears alongside the Reset conversation button.

6. **Click the green Download button** to save the file. Open the filled template and check the values: for example, view the `data_source` column on the Parameters sheet to confirm that values from your file are tagged `user_file` (not `ipcc_default`) in case there were any translation failures. When you are satisfied that the values in the template match your input data, proceed to the **1. Data Input** tab in the online interface.

When to use. The translator is most useful when (a) your data is in a different format from the template (e.g. rows are parameters, columns are years), (b) your column names differ from the template (e.g. `cattle_pop`, `live_wt_lb`), (c) you have data in inconsistent units (lbs vs kg, fractions vs percentages), or (d) you want IPCC defaults filled in for the parameters you do not measure.

Quality control. The translator does column-mapping, not analysis. Always (a) spot-check the key numbers (population, body weight, milk yield) by comparing with your original file, and (b) review the QA/QC tab after upload: any value the translator filled from an IPCC default is flagged in amber there.

Privacy + cost. The translator is hosted by the GMH project on a small AI account (no payment from you). The chat does not store your input data: only your email and a per-call token count are

logged so the administrator can monitor monthly usage. A $10/month global spending cap prevents runaway use; if the cap is hit you will see a clear "monthly cap reached, contact administrator" message.

2.3 Complete parameter reference

The table below lists all parameters accepted by the tool. Required parameters depend on the emission sources you are running. Parameters marked *n/a* in the IPCC default column must be supplied from your national data. The tool will auto-fill missing optional parameters with IPCC defaults and flag them in Tab 1 (see Section 3.2). When uploading your data, be sure to select 2006 IPCC Guidelines or 2019 Refinement so that the tool auto-fills and performs data quality checks according to the correct version of the guidelines.

Note: the table presents illustrative default values, not the complete range of defaults provided in the 2006 IPCC Guidelines or 2019 Refinement.

Table 1: Complete parameter reference: all Tier 2 inputs accepted by the tool.

Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Activity data						
N	Number of animals in this sub-category	head	*n/a*	±10%	Normal	Eq. 10.1
BW	Average live body weight (also TAM = Typical Animal Mass)	kg	270	±15%	Normal	Eq. 10.3 / 10.6
MW	Mature (adult) body weight: used in NEg growth equation	kg	300	±10%	Normal	Eq. 10.6
WG	Average daily weight gain: set 0 for non-growing adults	kg/day	0	±30%	PERT	Eq. 10.6
Milk	Average daily milk yield *per head*, calculated as total annual production divided by 365 (IPCC Vol.4 Ch.10). This is already averaged across the whole year including the dry period, so do **not** scale it by the lactating or calving share of the herd. Set 0 for sub-categories that do not produce milk.	kg/head/day	1.2	±20%	Normal	Eq. 10.8
Fat	Fat content of milk	%	4.3	±10%	Normal	Eq. 10.8

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
pct_pregnant	Fraction of females in this sub-category pregnant during the year, 0–1: includes pregnant heifers that have not yet calved. Aliases `pct_calving` and `pct_lactating` are accepted on upload, but the three are not the same quantity: this field weights C_p in Eq. 10.13 and the milk-N term in Eq. 10.33, and it does **not** weight lactation.	fraction	0.52	±20%	Beta	Annex 10A.1 (Africa, low productivity); Eq. 10.13
MilkPR	Protein content of milk: feeds the milk-N term in Eq. 10.33 (N retention rates for cattle, where the 6.38 milk-protein-to-N conversion is defined). The 2019 Refinement gives the relation %MilkPR = 1.9 + 0.4×%Fat (so 4.3% fat → 3.6% protein, the Africa value); illustrative values are tabulated in Annex 10A.1 / 10A.2 / 10A.3.	%	3.6	±10%	Normal	Eq. 10.33; Annex 10A.1–10A.3
Energy coefficients						
DE	Digestible energy as % of gross energy: typical range 45–75%	%	51.0	±15%	Normal	Eq. 10.14
Cfi	Maintenance energy coefficient: depends on sex and lactation status	MJ/day/kg$^{0.75}$	0.386	±30%	PERT	Table 10.4

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Ca	Activity coefficient for locomotion energy	dimensionless	0.17	±30%	Triangular	Table 10.5
C	Growth coefficient for NEg equation: depends on sex (female 0.8, castrate 1.0, bull 1.2)	dimensionless	0.8	±30%	Triangular	Eq. 10.6
Cp	Pregnancy coefficient: 0.10 for pregnant animals	dimensionless	0.10	±10%	Beta	Table 10.7
hours	Daily working hours: set 0 when animals do no work (typical for dairy/beef cattle inventories)	hours/day	0	±20%	PERT	Eq. 10.11
Tw	Mean daily temperature in winter (°C): Cfi cold-climate adjustment. Leave blank or set 20 to disable.	°C	20	±25%	Normal	Eq. 10.2
Enteric fermentation						
Ym	Methane conversion factor (% of gross energy in feed converted to CH_4). Depends on the animal class *and* on the guidelines edition, so the template fills it per sub-category rather than using one value: under the 2019 Refinement, 7.0 for non-dairy and multi-purpose cattle and 4.0 for feedlot cattle; under 2006, 6.5 for everything except feedlot at 3.0.	%	6.5	±20%	PERT	Table 10.12
Manure management: CH_4						

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
Bo	Maximum CH_4 producing capacity of manure	m^3 CH_4/kg VS	2019R: 0.13 (Other regions, low productivity cattle); 0.24 (intensive dairy NA/W.Europe); 2006 Africa = 0.10	±15%	PERT	Table 10.16(a) (Updated)
ASH	Ash content of manure (fraction of dry matter)	fraction	0.08	±25%	PERT	Eq. 10.24
UE	Urinary energy as fraction of gross energy	fraction	0.04	±25%	PERT	Eq. 10.24
CP	Crude protein content of diet: used to estimate nitrogen excretion	%	9.6	±15%	Normal	Eq. 10.32
Manure management: N_2O (indirect emission factors)						
EF4	N_2O EF for atmospheric N deposition (used in Eq. 10.27)	kg N_2O-N/(kg NH_3-N + NO_x-N)	**0.014** (2019R wet climate, the shipped default); 2019R aggregated 0.010; dry 0.005; 2006: 0.010	asymmetric	PERT	Eq. 10.27; Ch.11 Table 11.3
EF5	N_2O EF for N leaching/runoff (used in Eq. 10.29)	kg N_2O-N/kg N	2006: 0.0075; 2019R: 0.011	asymmetric	PERT	Eq. 10.29; Ch.11 Table 11.3
*The managed-storage N_2O quantities EF3(S), Frac_GasMS and Frac_LeachMS are entered **per manure-management system** in the Manure_Management tab (not in the Parameters tab), because each value is specific to a storage system. The tool reads them from there.*						
Manure management: N_2O (pasture/range/paddock)						

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Code	Definition	Unit	IPCC default	Suggested uncertainty	Suggested distribution	IPCC reference
EF3_PRP	N_2O emission factor for dung/urine deposited on pasture (used in Ch.11 Eq. 11.1)	kg N_2O-N/kg N	**0.006** (2019R $EF3_{PRP,CPP}$ wet climate, the shipped default); aggregated 0.004; dry 0.002; 2006: 0.02	asymmetric	PERT	Ch.11 Eq. 11.1; Table 11.1
Frac_GasPRP	Fraction of N volatilised from dung/urine on pasture (used in Ch.11 Eq. 11.9)	fraction	0.21 (cattle)	asymmetric	PERT	Ch.11 Eq. 11.9; 2019 Table 11.3
Frac_LeachPRP	Fraction of N lost through leaching from pasture deposition (used in Ch.11 Eq. 11.10)	fraction	2006: 0.30; 2019R: 0.24 (wet) / 0 (dry)	asymmetric	PERT	Ch.11 Eq. 11.10; 2019 Table 11.3

Note on asymmetric uncertainty: Parameters flagged "asymmetric" use explicit lower/upper bounds from the IPCC tables rather than a symmetric ±% formula. The template pre-fills these bounds automatically from the IPCC 2019 Refinement tables.

2.4 Understanding the input structure

The tool uses an Excel template with a specific structure. On the **Data Input** tab, under `Data source', select`Custom Upload', then select the IPCC Guidelines version, and download the template by clicking "Download blank template".

2.5 The Inventory_Metadata sheet

The first sheet captures where the inventory is for. Two cells deserve a note:

- **Country**: free text (e.g. `Zimbabwe`). Used in the report header, not used for any lookup.
- **Continental region**: a **dropdown** (`africa / asia / europe / americas / oceania / global`). This drives the QA tab's BW deviation check, which compares your body-weight value against the IPCC Vol.4 Ch.10 Annex Tables 10A.1 / 10A.2 / 10A.3 default for your continent. Pick the continent your inventory's animal sub-categories are in. If you genuinely do not know or are running a hypothetical example, select `global`.

The Parameters sheet has the following key columns:

Column	Description	Example
`cattle_type`	Broad cattle category	`dairy`, `non_dairy`
`aggregation_level`	Production system or region name	`Highland smallholder dairy`
`sub_category`	Animal sub-group	`cows`, `heifers`, `calves_female`
`parameter`	Parameter code (see Table 1)	`N`, `BW`, `Ym`, `DE`, `Bo`
`mean`	Central estimate (point value from your inventory)	`450` (BW in kg)
`distribution`	Uncertainty distribution shape	`normal`, `pert`, `lognormal`
`uncertainty_pct`	Symmetric ±% half-width of 95% CI	`20`
`lower`	Lower bound (alternative to uncertainty_pct)	`360`
`upper`	Upper bound	`540`
`param_type`	IPCC framing	`activity_data` or `coefficient`
`unit`	Unit of measurement	`kg`, `%`, `fraction`

Important: Each row is one parameter for one sub-category. A country with two production systems and five sub-categories each would have approximately 12–15 parameters × 10 sub-categories = 100–150 rows. (Hence, the AI translator may be particularly useful for users with large numbers of sub-categories.)

2.6 Defining sub-categories

Sub-categories should match your national inventory. Examples of typical sub-categories include:

- Dairy: `cows` (lactating), `dry_cows`, `heifers`, `calves`, `bulls`
- Non-dairy beef: `breeding_cows`, `growing_males`, `heifers`, `calves`, `oxen`

Each sub-category needs its own rows for each relevant parameter. The values of `cattle_type`, `aggregation_level`, and `sub_category` together identify each group uniquely: they must be spelled consistently across all rows for the same group. (If manually entering data, consistency in spelling can be achieved by 'dragging' an already-filled cell across unfilled cells, rather than retyping words.)

2.7 How many groups the tool can run

A **group** is one combination of `cattle_type`, `aggregation_level` and `sub_category`. An inventory with 20 regions and 9 animal sub-categories has 180 groups.

The hosted version of the tool runs on a single server with about 1 GB of memory, and each group costs roughly 1.8 MB while a simulation is running. That puts the practical ceiling at **about 400 groups**. Beyond it a run is more likely to stop partway through than to finish, and it stops with a memory error rather than a clear message.

If your file is over the limit the tool tells you when you upload it, not when you run it. The warning appears on the Data Input tab and says how many groups it found.

This is a limit of the hosted server, not of the method. If you run the tool locally in R, the only limit is your own machine.

2.7.1 Aggregating your data

The fix is to describe the same herd at a coarser level before you upload. A national inventory reported by department rather than by municipality is the usual case.

To combine several fine groups into one coarse group:

- **Populations are added together.** The coarse group's `N` is the sum of the fine groups' `N`.
- **Every other parameter is a population-weighted average.** For body weight, milk yield, digestibility and the rest, multiply each fine group's value by its population, add those up, and divide by the total population.
- **Manure system shares are population-weighted in the same way**, and should still sum to 100 per cent within each coarse group.

Aggregating is not a compromise forced on you by the server. The IPCC recommends it where sub-categories are fully correlated, which is exactly the case when they share the same emission factors (2019 Refinement, Volume 1, Chapter 3). A department-level inventory is also the resolution at which most countries report.

2.7.2 Do not split the file and run the parts separately

It is tempting to cut a large inventory into several files, run each one, and add the results. **The totals would be right and the uncertainty would be wrong.**

Adding the means of the parts is correct: emissions are additive. Combining their confidence intervals is not, because the parts are not independent. They share the same IPCC coefficients, so if the true value of a methane conversion factor is higher than assumed, every part is wrong in the same direction at the same time. Treating the parts as independent and combining their intervals in the usual way makes the total uncertainty look far smaller than it is, and the more pieces you cut the file into, the smaller and the more wrong it looks.

There is no correct way to reassemble the confidence interval outside the tool. Aggregate the inputs and run once.

2.7.3 If you genuinely need the fine detail

Two options, in order of effort:

- **Run the tool locally.** It is open source, and on a desktop machine with 16 GB of memory a few thousand groups are no problem.
- **Run the coarse inventory for reporting, and fine subsets for diagnosis.** For example, run all departments once for the national figure, then run a single department at municipality level to see where the uncertainty within that department comes from. Report the first; use the second to decide where better data is worth collecting.

2.8 If you already measured feed intake

Most inventories let the tool work out how much each animal eats, from body weight, growth, milk yield, pregnancy and activity. If your own data already measures intake, you can supply it instead and skip that step.

Two optional rows appear for every sub-category in the Parameters sheet:

- `GE`: gross energy intake, in MJ per head per day.
- `DMI`: dry matter intake, in kg per head per day. Used only when `GE` is not given, and converted at the IPCC factor of 18.45 MJ per kg.

Leave them blank unless you have a measurement. A blank cell means "calculate it from the animal performance data", which is the normal case and exactly what earlier versions of the tool did. Fill them in only for the sub-categories where you actually have a figure; you can mix measured and calculated groups in the same inventory. If you enter both, `GE` is used, because it carries your own measured energy density rather than the IPCC average.

You still need milk yield, weight gain and digestibility. This surprises people, so it is worth being explicit. A measured gross energy replaces the energy calculation, not the nitrogen calculation. Nitrogen excretion works by subtracting the nitrogen retained in milk and in weight gain from the

nitrogen eaten, so those inputs still drive your nitrous oxide numbers. Digestibility (`DE`) is still needed too: once you supply `GE` it no longer affects enteric methane, but it still decides how much of the feed passes through undigested and becomes manure, and therefore manure methane.

Checking which route was used. The results table has an **Intake route** column showing, for each group, whether the figure was calculated ("Energy balance") or supplied ("Measured GE" or "Measured DMI"). The same information appears in the run settings of the Excel download and in the Word report, so anyone reviewing your inventory can see how each number was produced. If a group mixes both, it is reported as "Mixed" rather than being labelled with one method.

2.9 Obtaining uncertainty estimates

Refer to the guidance in the IPCC Guidelines Vol.1 Ch.3. In brief:

For activity data (N, BW, milk yields):

- **Survey-based estimates**: If populations are estimated from representative livestock surveys, the survey sampling error may provide a direct estimate of uncertainty. The 95% confidence interval from a survey translates directly to the `lower` and `upper` bounds (or to `uncertainty_pct` if symmetric).
- **Expert elicitation**: Ask the data provider to give a range within which they are 95% confident the true value falls.
- **IPCC defaults**: IPCC 2019 Refinement Chapter 11 provides uncertainty ranges for key emission factors. Use these as the starting point if no country-specific data is available.

For emission factors and coefficients (Ym, Bo, DE, MCF, etc.):

- **Literature review**: If you used a country-specific value from a published study, the study's reported confidence interval or standard deviation can be used.
- **IPCC default uncertainty ranges**: IPCC 2019 Refinement (Vol.4 Ch.10 / Ch.11) gives uncertainty percentages or lower / upper bounds for several parameters (e.g. $EF_{3,PRP}$, Ym, MCF). The template pre-fills these where IPCC publishes them; where IPCC does not publish an uncertainty range, the template falls back to ranges from Penman *et al.* (2000) and Monni *et al.* (2007), these are clearly distinguished in the template footnotes.
- **Expert elicitation**: Ask the expert who provided the value to estimate the plausible range.

If you genuinely have no information about a parameter's uncertainty, a reasonable default for emission factors is ±20–40%. For population data with census-based estimates, ±5–15% is typical.

2.10 Choosing a distribution

Distribution	Shape	Best for
Normal	Symmetric bell curve	Activity data (N, BW) with symmetric measurement errors
PERT	Smooth bounded curve	Emission factors with a min, mode, and max
Triangular	Peaked triangle	Clear most-likely value with known hard limits
Uniform	Flat (equal probability)	Conservative fallback when only bounds are known
Lognormal	Right-skewed	Strictly positive values with right-skewed uncertainty (Bo, EFs)
Beta	Bounded $[0, 1]$	Fractions (pct_calving, Cp)

Recommended defaults:

- Use **Normal** for population (N) and body weight (BW): typically measured with roughly symmetric errors
- Use **PERT or Lognormal** for emission factors (Ym, Bo, MCF, EF3–EF5, Frac parameters): bounded below at zero with right-skewed uncertainty
- Use **Beta** for fractions bounded to [0, 1] (pct_calving, Cp)
- Use **Uniform** as a conservative fallback when you only know the plausible range

3 Loading your data (Tab 1: Data Input)

3.1 Using the example datasets

Two built-in example datasets are included to let you explore the tool without uploading your own data:

- **Country X (hypothetical dairy)**: A fictional East African highland dairy system, 12 parameters, dairy/cows, including a synthetic 5-year time series.
- **Country Y (hypothetical pastoral)**: A fictional semi-arid pastoral beef system, 11 parameters, non-dairy/breeding cows, including a synthetic 5-year time series.

Select either from the "Country / Example Data" dropdown. The tool loads the data immediately and you can proceed directly to Tab 2.

3.2 Uploading your own data

Step 1: Choose an IPCC Guidelines version. Before downloading the template, select the IPCC version your inventory follows:

- **IPCC 2006**: Uses the eight original manure management systems (MMS). Choose this if your national inventory is based on the 2006 Guidelines.
- **IPCC 2019 Refinement**: Adds four additional MMS types (anaerobic digester/biogas, aerobic treatment, solid storage covered/compacted, burned for fuel) and uses updated emission factors and N excretion equations from the 2019 Refinement. Choose this for inventories aligned with the 2019 Refinement.

The MMS dropdown in the downloaded template is automatically filtered to show only the manure systems valid for the version you selected.

Step 2: Download and fill in the template.

1. Click "Download Excel template" to get the pre-formatted workbook
2. Fill in the **Parameters** sheet following the structure in Section 2.3
3. Fill in the **Manure_Management** sheet if you are running manure N_2O sources
4. *(Optional)* Fill in the **Parameter_TimeSeries** sheet if you want correlations computed automatically from historical data
5. Save the file as .xlsx and upload it using the file picker.

Step 3: Review upload notifications.

After a successful upload, a green notification confirms how many parameters were loaded. Any warnings appear as amber notifications.

Missing parameters and auto-fill: If your upload does not include all required parameters, the tool will:

- Still load and display all parameters that *were* supplied: you can review them immediately in

Tab 2 and Tab 3.

- Automatically fill in missing parameters using IPCC default values (see the "IPCC default" column in Table 1).
- Show a **yellow warning notification** listing exactly which parameters were auto-filled and with what values (e.g. "*3 parameters auto-filled from IPCC defaults: Ym = 6.5, Bo = 0.13, DE = 51.0*").
- Flag those parameters as **Missing** (purple/grey) in the QA/QC tab (Tab 2).

You should review whether the IPCC defaults are appropriate for your country context and, where necessary, revise any auto-filled values before your final simulation run.

Common upload issues:

- `cattle_type` must be `dairy` or `non_dairy` (lowercase, no spaces)
- Distribution names must match exactly: `normal`, `pert`, `triangular`, `uniform`, `lognormal`
- If `uncertainty_pct` is left blank, both `lower` and `upper` must be filled
- `param_type` must be `activity_data` or `coefficient`

4 Reviewing data quality (Tab 2: QA/QC)

4.1 What the QA/QC checks do

The QA/QC tab runs automated checks on your parameter values and flags any that are potentially problematic. Five statuses are possible:

Status	Colour	Meaning
Pass	Green	All structural checks pass; for `BW`, the mean is also within ±50% of the continental IPCC default (Annex Tables 10A.1 / 10A.2 / 10A.3).
Warn	Amber	For `BW`: the value deviates more than 50% from the continental IPCC default, check and document (above 200% the message is stronger, but it is still advisory). For any parameter: a non-fatal structural condition fired (e.g. implausible bounds $BW > 1500$ kg or $Y_m > 12\%$, fractional parameter with an unbounded distribution, asymmetric-bounds reminder for $EF_{3,PRP}$ / Frac_GASM_PRP / Frac_LEACH_PRP, sub-category auto-match).
Fail	Red	$DE_{\%} \leq 0$ or ≥ 100; bounds that do not bracket the mean; an MMS fraction-sum more than 1% off 100%; or a Beta configured with mean outside $(0, 1)$. A fail blocks the run until it is fixed or explicitly ignored (see below).
Missing	*Grey*	Parameter not found in upload; auto-filled with the IPCC default from PARAM_CATALOGUE (Vol.4 Ch.10 / Ch.11). Applies to every parameter.

Scope of the benchmark-deviation check. The deviation check (Pass below ±50% from the IPCC default, Warn above; a deviation above 200% is still a warning, never a fail) is applied **only to BW (body weight)**: the one parameter for which the tool keeps a defensible continental IPCC table lookup: Annex Tables 10A.1 (dairy cows), 10A.2 (non-dairy cattle), and 10A.3 (buffalo).

Cross-sheet sub-category-key check. If a Parameters-sheet `sub_category` (e.g. `DINT_heif`) has no exact match in the Manure_Management sheet but *is* close to exactly one Manure_Management sub-category (e.g. `DINT_heifer`, edit distance ≤ 2), the tool auto-matches them and emits a *Warn* row labelled `sub_category_auto_match` so you can verify the substitution. If multiple candidates match, the run is blocked with a clear error.

4.2 What to do for each status

Pass: No action needed. All structural checks passed; for `BW` the mean is also within ±50% of the continental IPCC default (Annex Tables 10A.1 / 10A.2 / 10A.3).

Warn: Either the `BW` mean deviates more than 50% from the continental IPCC default, or a

non-fatal structural condition fired (asymmetric-bounds reminder for $EF_{3,PRP}$ / Frac_GASM_PRP / Frac_LEACH_PRP, sub-category auto-match, implausible bounds, fractional parameter on an unbounded distribution). Possible actions:

- Confirm the value matches your country context.
- If it is country-specific and well-justified, document the source in your inventory report.
- If you intended to use a climate-disaggregated default, check IPCC Vol.4 Ch.11 Tables 11.1 / 11.3 for the wet/dry value.
- If it is an error, correct it in your Excel file and re-upload.

Fail: A structural validation failed: $DE_{\%} \leq 0$ or ≥ 100; bounds that do not bracket the mean; an MMS fraction-sum more than 1% off 100%; or a Beta configured with mean outside $(0, 1)$. The Run button stays blocked while any fail remains. Correct the value in your Excel file and re-upload, or, if you have checked the value and accept it, tick **Ignore** next to the row.

Ignoring a failure. Each failing row has an Ignore checkbox. The first tick opens a warning: running with ignored checks may produce unreliable results. Two failures cannot simply be waved through because the sampler would return NaN, so ignoring them applies the smallest repair instead and marks the row *Repaired*: bounds that do not bracket the mean are widened around it, and a Beta or log-normal distribution on a value outside its valid range is replaced by a normal distribution. An ambiguous sub-category key cannot be ignored. Every ignored or repaired check is shown in a red banner above the results and listed in the Excel report (sheet `QA_Caveats`), at the end of the CSV, and in section 2b of the Word summary, so a reviewer always sees them next to the numbers.

Missing: A required parameter was not found in the upload. The tool auto-filled it with the IPCC default value from PARAM_CATALOGUE; the value and the IPCC equation / table source are shown in the Tab 1 notification. Applies to every parameter, not just `BW`. Review whether the default is appropriate for your country context and, where possible, replace it with country-specific data.

Tip: You can make any necessary corrections by editing in Tab 3 (`Uncertainty'). The final values used in the simulation runs will be documented in the reports output by the tool, but will not be updated in your data input file. Alternatively, if you edit the values in your data input file and re-upload in Tab 1` (Data input'), the values used will be documented in the final reports and stored in your data input file.

4.3 Documenting deviations

For any Warn or Info flags where you are deliberately using a country-specific value that deviates from the default, document the justification in your national inventory report. IPCC expert reviewers will look for this documentation.

5 Reviewing uncertainty distributions (Tab 3: Uncertainty)

5.1 The parameter table

Tab 3 shows all parameters loaded from your data as an editable table. Each row shows:

- The parameter name, unit, and current mean value
- The distribution type currently assigned
- The uncertainty_pct or lower/upper bounds
- A visual preview of the distribution shape

You can edit values directly in the table. Clicking a cell opens an edit field.

5.2 Quick-set buttons

Two quick-set buttons are available for rapid exploration:

- **Set all activity data to Normal ±15%**: Sets N, BW, and other activity data parameters to a Normal distribution with ±15% uncertainty. Click again to undo.
- **Set all coefficients to PERT**: Sets emission factors and other coefficients to PERT distribution. Click again to undo.

These are useful for a first exploratory run. Refine individual parameters before your final analysis.

6 Correlations (Tab 4: Correlations)

Deeper dive. For a side-by-side worked comparison of all four correlation modes applied to the Country X / Country Y built-in examples (including how each mode changes the 95% confidence interval on total CO_2eq, the structural-defaults heatmap with per-pair sources, and the common mistakes to avoid), click the green *Find out more* link at the top of the Activity Data Correlations card on Tab 4. It opens a focused topic page in a new browser tab.

6.1 Why correlations matter

There are two different things called "correlation" on this page, and they matter to very different degrees.

Between parameters, within a group. If two parameters tend to move together (e.g. larger animals also tend to produce more milk), ignoring the relationship understates the uncertainty in the total. For most Tier 2 analyses this is a modest effect, typically a few percentage points, and ignoring it is a reasonable simplifying assumption. IPCC Approach 2 guidance recommends including known correlations where they exist.

Between groups, for the same parameter. This one is not modest. If several of your groups carry the *same* number for a coefficient, because it came from one IPCC default or one regional study, then that is a single estimate applied many times. If it is wrong, every group is wrong in the same direction at once, and nothing cancels. Sampling each group independently instead makes the errors average each other out, and the reported uncertainty falls as the inventory is split more finely.

The size of this effect is easy to underestimate. On a test inventory where the herd, the parameters and the uncertainty inputs were all held fixed and *only* the number of groups changed, the reported 95% margin of error fell from 25.1% on a single group to 1.5% on 251 groups. Nothing about the data had changed; only the disaggregation had. This is why **Inputs shared between groups** defaults to treating a shared value as one estimate, and why you should normally leave it there. IPCC 2019 Refinement Vol.1 Ch.3 makes the same point: subcategories "can be highly correlated, because either the AD are derived from the same source or the EFs have parameters in common".

6.2 Inputs shared between groups

This setting sits below the correlation heatmap and has two options.

Treat a shared value as one estimate (the default, and the right answer for nearly every inventory). Where several groups carry the same number for a coefficient, the tool draws it once and applies that draw to all of them. Their errors then move together, which is what actually happens when one estimate is reused.

Treat every group independently. Each group redraws every coefficient on its own. This is correct only when each group's values really were estimated separately, which is unusual. It is kept so you can reproduce a previous run or see the size of the effect on your own data.

Underneath the control, the tool tells you what it found, for example *"14 coefficients are used identically across groups in this inventory (Ym, Bo, EF3, ...)"*. If that list is empty, nothing in your inventory is shared and the setting changes nothing.

Two things to expect. First, the reported margin of error will usually be **larger** with the default than with independent groups. That is the correction, not a regression: the smaller figure was an artefact of splitting the inventory. Second, the tool decides what is shared by looking for identical values, so if you genuinely have a separate estimate per group, enter your own numbers and those groups will be treated independently automatically.

6.3 Quick guide: which option do I pick?

- Do you have ≥ 5 years of national time-series data in your upload? → **From template (auto, time-series)**.
- No time series, but you want to capture well-known biological linkages (BW ↔ MW, Milk ↔ Fat, DE ↔ Ym, ...)? → **Structural defaults (literature-based)**. This is the recommended pick for single-year inventories without historical data.
- You have your own correlation matrix from a study or expert elicitation? → **Advanced: manual entry**.
- None of the above? → leave **No correlations** (the default; conservative and defensible).

Options you can't pick are greyed out. If a mode would have nothing to work with (e.g. `From template'' when your \texttt{Parameter\_TimeSeries} sheet is empty, or`Advanced: manual entry'' before you've uploaded a CSV), the radio choice appears greyed out with a short note explaining why. The tool also blocks the Run button with an explicit error if a no-data mode is somehow selected.

6.4 What change in the result should I expect?

The central emission estimate is essentially unchanged regardless of which mode you pick: correlations preserve marginal means by construction. What changes is the **width of the 95% uncertainty interval**. As a rule of thumb, on a typical single-year cattle inventory:

- **No correlations** → the baseline. All parameters drawn independently.
- **Structural defaults** → changes the 95% uncertainty interval by **about one percentage point or less** on the built-in examples (Country X 25.3% → 25.2%; Country Y 36.5% → 37.5%). The effect is modest because the preset links only 7 pairs out of ∼78 possible, with values between 0.20 and 0.50 in absolute terms. Strong correlations on wide inputs, entered through the manual matrix, move the interval much more.
- **From template (auto, time-series)** → varies with the data. A rich, varied historical series (multiple parameters genuinely co-moving year-on-year) can produce effects similar to or larger than the structural defaults. A sparse or mostly static series will produce a much smaller effect

because the matrix collapses to near-identity once constant columns are dropped.

- **Advanced: manual entry** → entirely depends on what you upload. Strong correlations on parameters with wide uncertainty bounds can move the interval by 20% or more (in either direction).

If you select correlations and the result barely changes, that is usually expected, not a bug: the IPCC equation chain multiplies around 13 parameters together, which dilutes the effect of any single correlated pair. See the Methodology document for the underlying mathematics.

Note: Before running your simulation, Tab 5 ('Simulate & Results') lets you select *compare with/without correlations*. Selecting this lets you inspect the influence of the correlation options you chose.

6.5 Automatic correlations from time-series data

If you included a Parameter_TimeSeries sheet in your upload, the tool can compute historical correlations automatically:

1. On Tab 4, select **From template (auto, time-series)** under Activity Data Correlations.
2. The tool computes the **Spearman** rank correlation matrix of the historical parameter data after first-difference detrending (year-on-year changes, which strips out shared long-run growth) and samples using **Iman–Conover restricted pairing**: the method explicitly cited by IPCC Vol.1 Ch.3 §3.2.3.2 (p. 26). Iman–Conover is distribution-free and reproduces the target rank correlation exactly, regardless of the marginal shapes.
3. Inspect the heatmap that appears below the controls to confirm correlations are plausible.

6.6 Manual correlation matrix

If you prefer to specify correlations manually:

1. Select **Advanced: manual entry** under Activity Data Correlations.
2. Upload a square, symmetric CSV matrix of correlation coefficients (-1 to $+1$, diagonal $= 1$). The value you enter for cell (i, j) is the **Spearman rank correlation** between parameters i and j, which the tool preserves through Monte Carlo per IPCC Vol.1 Ch.3 §3.2.3.2.
3. The tool verifies positive-definiteness and projects onto the nearest positive-definite correlation matrix if needed; you are warned if any entry shifts by more than 0.01.

Downloadable starting templates. Two CSV templates appear inside the **Advanced: manual entry** panel and remove the need to construct the matrix from scratch:

- **Download blank matrix template**: a square matrix with all parameter names already in place as row and column headers, the diagonal set to 1, and every off-diagonal cell set to 0. Open it in Excel, fill the cells you want correlated, save as CSV, and upload via the **Upload correlation matrix (.csv)** button.

- **Download matrix with example values**: the same matrix but pre-filled with the seven structural-defaults pairs (BW ↔ MW, BW ↔ WG, Milk ↔ Fat, Milk ↔ BW, Milk ↔ DE, DE ↔ CP, DE ↔ Ym). This matrix is **identical** to what the in-app *Structural defaults* preset applies internally; it is a useful starting point if you want to tweak just one or two pairs.

Parameter-name rules. The templates ship with the canonical parameter names already in the first column and the header row (`N`, `BW`, `MW`, `WG`, `Milk`, `Fat`, `pct_pregnant`, `DE`, `Cfi`, `Ca`, `C`, `Cp`, `hours`, `CP`, `Ym`, `Bo`, `ASH`, `UE`, `EF3_PRP`, `EF4`, `EF5`, `Frac_GASM_PRP`, `Frac_LEACH_PRP`, `MilkPR`, `Tw`). **Do not rename rows or columns**: names that don't match a catalogue parameter are silently ignored when the matrix is applied to the simulation (no error message; the correlation just doesn't take effect).

Missing parameters are assumed to have zero correlation. If your uploaded matrix omits a parameter that the simulation uses (for example, you upload a small matrix covering only the parameters you want correlated), every pair involving the omitted parameter is treated as independent: the simulation runs and uses the correlations you did provide, but it does not invent values for the missing ones. Equivalent rule for header typos: a misspelled row/column name is treated the same as a missing parameter, so the correlation simply does not take effect. This is intentional graceful degradation, not an error, so the easiest way to verify your matrix is being used as you expect is to use the downloadable templates above (which ship with all canonical names already in place).

6.7 Structural defaults (literature-based)

A built-in preset of structural defaults is available. It applies the seven pairs below, drawn from the IPCC equations and the livestock literature. *The values are literature-based: the direction of each link comes from the IPCC equations and published studies, and the size was set by the project team from the evidence listed; the IPCC publishes no numerical correlation coefficients for these pairs.* Hover any non-zero cell in the heatmap for the per-pair source.

Parameter A	Parameter B	ρ	Biological reason and references
BW	MW	+0.50	Live weight and mature weight enter the IPCC net-energy-for-growth equation together, and animals of larger breeds are heavier at every age and have a larger mature weight. Set below the same-survey value because MW is usually a breed-reference constant while BW comes from the census. IPCC 2006 Eq 10.6; IPCC 2019 Table 10A.2.
BW	WG	+0.40	Live weight and daily gain enter the same equation, and well-fed animals are both heavier and gain faster; the sign can reverse for young stock approaching mature size. IPCC 2006 Eq 10.6; Mehrban et al. 2021 (Animal Bioscience 34:824–832); IPCC 2019 Table 10A.2.
Milk	Fat	−0.30	Dilution effect: as milk yield rises, fat yield rises less than proportionally, so the fat percentage falls. Boujenane 2002 (Rev. Elev. Med. Vet. Pays Trop. 55:63); Nistor et al. 2011 (Sci. Pap. Anim. Sci. Biotechnol. 44:316); IPCC 2019 Table 10A.1.
Milk	BW	+0.30	Larger dairy breeds and more intensive systems have both heavier cows and higher milk yields; within one breed the link is weak. IPCC 2019 Table 10A.1; Berry and Evans 2022 (JDS Communications 3:32–37); Berry et al. 2003 (J. Dairy Sci. 86:2193–2204).
Milk	DE	+0.30	More digestible feed gives more net energy per kg of feed and supports higher milk yield. NRC 2001 Ch. 2 (Energy); IPCC 2019 Table 10A.1.
DE	CP	+0.20	Within one forage, digestibility and crude protein fall together as the plant matures; across different feeds the link is weak because protein and digestibility depend on different things (plant family, plant part, processing). INRAE-CIRAD-AFZ feed tables 2018 (feedtables.com); Feedipedia.
DE	Ym	−0.20	IPCC Table 10.12 assigns a lower Ym to more digestible diets, mainly because high digestibility goes with high feeding level and grain-rich rations; within forage-based diets, methane per unit of energy changes little with digestibility or rises slightly. IPCC 2019 Table 10.12; Ramin and Huhtanen 2013 (J. Dairy Sci. 96:2476–2493); Johnson and Johnson 1995 (J. Anim. Sci. 73:2483–2492).

All other pairs are set to zero. Four of the values can reasonably be changed through the manual matrix when you know your situation: BW ↔ MW 0.8–0.9 when both weights come from the same survey; Milk ↔ BW 0.10–0.20 for a single-breed herd; DE ↔ CP 0.3–0.5 when your feed-quality uncertainty is mainly about forage maturity; DE ↔ Ym −0.3 to −0.5 for high-grain dairy or feedlot systems.

7 Running the simulation (Tab 5: Simulate)

7.1 Settings

Before clicking Run, configure:

Number of iterations: The number of Monte Carlo samples.

- **1,000**: Quick test run to check the model runs correctly. Do not use for reporting.
- **10,000**: Minimum for final reporting purposes. Tail percentiles (2.5th/97.5th) are reliable.
- **25,000–30,000**: Recommended for final reporting, especially when correlations are enabled or you have many sub-categories.

GWP version: Choose between AR4 (CH_4 GWP = 25, N_2O = 298), AR5 (CH_4 GWP = 28, N_2O = 265) and AR6 (CH_4 GWP = 27, N_2O = 273; the 27 is the non-fossil/biogenic methane value, appropriate for cattle). The default is AR5. Select the version required by your national inventory reporting guidelines.

Random seed: Fixing the seed ensures results are exactly reproducible. Change the seed to verify convergence: if the 95% CI changes substantially between runs, increase the number of iterations.

Analysis mode: Single-year (e.g. simulation run only on the latest year in your inventory time series) or Trend (multi-year time series).

Emission sources: Tick all sources applicable to your inventory. Unticked sources contribute zero emissions to the total. (Note: you can choose not to select enteric fermentation, but the Tier 2 input data for enteric fermentation should be supplied. Not selecting enteric fermentation means that the emissions and uncertainty of this emission source will not be reported in the results.)

7.2 Trend mode: year-to-year coefficient correlation

If you picked **Trend** analysis mode on Tab 1, an extra control appears on Tab 5: **Year-to-year correlation**. It governs how the IPCC coefficients (Cfi, Ca, Ym, Bo, MCF, EF3, EF4, EF5, ...) move from one year to the next *within a single Monte Carlo iteration*. Activity data (N, BW, Milk, ...) is always re-sampled fresh each year, this setting only affects the per-head coefficients.

Why does this matter? If your country uses the same emission factors for the whole inventory time series (e.g. when Tier 2 EFs were estimated using one dataset and applied to all inventory years), then a high-Ym draw in your base year should also be a high-Ym draw in the current year. The coefficient uncertainty then cancels when you compare the two years, so the trend uncertainty reflects only the activity-data changes. This is what IPCC Vol.1 Ch.3 §3.2.3 calls a "Type A'' sensitivity. If you instead treated EFs as independent across years, you would artificially inflate the trend uncertainty.

The three options:

- **Fully correlated coefficients (IPCC 2019 default)**: the same coefficient draw is reused for

every year of the inventory within one MC iteration. *Pick this if your EFs are IPCC defaults or come from a single national estimation programme reused across the whole series.* This is the IPCC-recommended default and the conservative choice for trend uncertainty.

- **Partial (AR(1), $\rho = 0.7$)**: coefficient draws drift slowly between years (last year's value gets 70% weight, a fresh draw 30%). *Pick this if your EFs are re-estimated periodically: e.g. every 3–5 years, and neighbouring years share most of the same observational basis.*
- **Independent (no year-to-year correlation)**: coefficient draws are sampled fresh each year. Maximises the EF contribution to trend uncertainty. *Pick this only if you genuinely re-measured every emission factor every year with fully new field data*, which is rarely realistic for a national inventory.

A practical sanity check: re-run your trend with `Fully correlated'' and again with`Independent''. If the two trend uncertainties are similar, your result is robust to this choice. If they are very different, your trend uncertainty is dominated by the EF block and you need to be explicit about which assumption you're making in your inventory documentation.

Trend mode runs n_iter simulations **per year**, so total compute = n_iter × number of years. Use 5{,}000–10{,}000 iterations for trend runs in development and 25{,}000–30{,}000 for final reporting.

7.3 Running the simulation

Click **Run Monte Carlo Simulation**. A progress bar shows the simulation steps. For 10,000 iterations and a single production system, typical run time is 15–60 seconds, but may be longer for inventories with large numbers of sub-categories.

When complete, results appear in the lower section of the same tab.

If something is missing or not working: All settings (iterations, GWP, seed, emission sources) remain active in the upper section of Tab 5, simply adjust any setting and click **Run Monte Carlo Simulation** again. The previous results are replaced automatically. If you need to change your input data (parameters, distributions, or correlations), go back to the relevant tab (Tab 1–4), make the change, then return to Tab 5 and re-run.

7.4 Convergence diagnostics

Click "Run Diagnostic" (magnifying glass button) after a simulation to assess whether enough iterations were run for stable results:

- **MCSE%** (Monte Carlo Standard Error %): Pass $< 0.5\%$, Warn $0.5-1\%$, Fail $\geq 1\%$
- **Mean drift %**: How much the mean from the first half of iterations differs from the second half. Pass $< 2\%$
- **CI drift %**: How much the 95% CI bounds shift between the two halves. Pass $< 5\%$
- **Convergence trace**: Running mean and 95% CI plotted against iteration count, should

stabilise before the end of the run

Note: this tool uses **independent Monte Carlo**, not MCMC. Each iteration is drawn independently, there are no chains, no warmup, and no burn-in.

8 Understanding results (Tab 5: Results)

8.1 Headline metrics

After a successful run, five value boxes show:

- **Total CO_2eq (t)**: The mean of the simulated total emission estimate, in tonnes CO_2-equivalent
- **Enteric CH_4 (t)**: Mean enteric fermentation contribution *plus* the source-specific 95% margin of error inline (e.g. `2,608 t · ±12%`)
- **Manure CH_4 (t)**: Mean manure CH_4 *plus* its 95% MoE
- **Pasture N_2O (t)**: Mean combined direct and indirect N_2O from pasture deposition *plus* its 95% MoE
- **Total 95% MoE (%)**: The headline uncertainty, half-width of the 95% confidence interval as a percentage of the mean. This is the IPCC reporting convention (Vol.1 Ch.3 Table 3.3) and the single uncertainty metric used across every user-facing display in the app (results tables, comparison chart, IPCC Annex 7, Word report).

Headline by cattle type. Directly below the value boxes, a "Headline by cattle type" card appears whenever your inventory contains more than one `cattle_type` on the Parameters sheet (typically `dairy` and `other`). It shows one row per cattle type with mean ± 95% MoE for each of the four headline sources plus the total CO_2eq. Single-cattle-type runs do not see this card.

8.2 Uncertainty decomposition (AD vs EF)

If "Run uncertainty decomposition (AD/EF/Combined)" was checked, the results include a bar chart with the 95% MoE on the y-axis (IPCC reporting convention) under three configurations:

- **AD uncertainty**: Driven by activity data only (N, BW, Milk, WG) with all emission factors fixed at their means
- **EF uncertainty**: Driven by emission factors and coefficients only with activity data fixed at means
- **Combined uncertainty**: Both sources varying together

If AD uncertainty dominates, improving livestock population surveys would most effectively reduce total uncertainty. If EF uncertainty dominates, investing in country-specific EF measurement studies would be more impactful.

8.3 By-system and by-category breakdown

- **By-system breakdown**: Mean, 95% CI, MoE%, and CV% for each production system and emission source. The aggregation-level selector at the top of the section lets you switch between `cattle_type` (default, dairy / other), `aggregation_level` (production system) and `sub_category`.

- **By reporting category (IPCC Table 3.3 layout)**: Grouped by IPCC source category (3.A.1, 3.B.1, 3.B.5, 3.D.1.c, 3.D.2). The leftmost column is always `Cattle type`, regardless of the aggregation-level selector. Each row carries mean values for the raw gas (`Mean (t` CH_4`)` *or* `Mean (t` N_2O`)`, only one is populated per source) *plus* `Mean (t` CO_2`eq)` for cross-source comparison, plus MoE%, CV%, and the 95% CI bounds.

9 Sensitivity analysis (Tab 6)

9.1 Reading the tornado chart

The tornado chart ranks parameters by their contribution to total uncertainty. The longest bars indicate the parameters that, if better constrained, would most reduce the 95% CI.

Each bar is labelled as `<parameter> (<sub_category>)`: for example `Ym (DINT_cow)` vs `Ym (DINT_heif)`. The sub-category in parentheses identifies which animal group the influential parameter belongs to, so you can target research at the specific sub-category driving uncertainty rather than at the parameter in general. The same labels appear in the rank-correlation table, the Excel SRC / PRCC sheets, and the Word report's sensitivity section. Per-MMS sampled parameters carry their MMS in the parameter name and the sub-category in parentheses, e.g. `MCF_solid_storage (DINT_cow)` or `fraction_pasture (DINT_heif)`.

Per-source sensitivity. The "Output" dropdown above the tornado lets you re-rank parameters for a single emission source (Enteric CH_4, Manure CH_4, etc.). The per-source view aggregates across ALL sub-categories: earlier versions of the tool silently restricted the per-source view to the first sub-category only, dropping every other sub-category's contribution; that has been fixed.

Bars are coloured by reducibility:

- **Green**: Uncertainty can be reduced with better local data (surveys or measurement studies)
- **Grey**: Uncertainty is largely irreducible with national-level data (e.g. fundamental biochemical constants or IPCC defaults that are rarely measured)

9.2 SRC vs PRCC

Two sensitivity methods are computed:

- **SRC** (Standardised Regression Coefficient): Linear sensitivity. Reliable when relationships are approximately linear.
- **PRCC** (Partial Rank Correlation Coefficient): Rank-based, captures non-linear relationships. More robust but less interpretable.

Both are shown; they typically agree on the top-ranked parameters. If they disagree significantly, the emission pathway may be strongly non-linear.

10 Generating the IPCC report (Tab 7: IPCC Report)

10.1 IPCC Table 3.3

The IPCC Report tab shows your results formatted as IPCC 2006 Vol. 1 Ch. 3 Table 3.3. The three uncertainty columns in the downloadable Excel sheet are:

- **AD uncertainty (95% MoE %)**: % uncertainty from activity data alone
- **EF uncertainty (95% MoE %)**: % uncertainty from emission factors alone
- **Combined uncertainty (95% MoE %)**: Overall % uncertainty

Definition of % uncertainty: Following IPCC Table 3.3 conventions, all values are expressed as the half-width of the 95% confidence interval divided by the mean, multiplied by 100, i.e. `MoE% = ((CI_upper - CI_lower) / 2) / mean * 100`.

Example: a Combined uncertainty of 35% means the 95% CI is approximately ±35% of the mean emission estimate.

10.2 Downloads

Three download formats are available:

- **Excel report (.xlsx)**: Full workbook with run settings, IPCC summary table, uncertainty metrics, and sensitivity results (SRC and PRCC)
- **CSV report**: The IPCC Table 3.3 formatted table only, for easy import into inventory reporting software
- **Word report**: Narrative summary suitable for inclusion in the uncertainty section of your national inventory report

11 IPCC reporting conventions

11.1 Where uncertainty results go in the national inventory report

IPCC 2006 Vol. 1 Ch. 3 §3.3 describes the standard approach for Approach 2 (Monte Carlo):

1. Report the 95% confidence interval for each major source category and for the national total
2. Present the results in Table 3.3 format (included in the IPCC Report tab downloads)
3. Describe the Monte Carlo approach used (number of iterations, distributions chosen, correlation assumptions)
4. Document any parameters where country-specific values were used instead of IPCC defaults, including the justification

The methodology PDF (available on the Resources tab) provides a complete technical description of the equations, sampling approach, and convergence assessment that can be referenced in your national inventory report.

11.2 What IPCC expert reviewers will check

Based on experience with IPCC expert reviews, reviewers typically focus on:

- Whether the uncertainty metric is reported at the appropriate level of disaggregation (e.g. separately reported for 'dairy' and 'other cattle' per emission source when emissions from these cattle types are separately reported in the inventory)
- Whether the IPCC Table 3.3 format and column conventions are followed
- Whether Approach 2 (Monte Carlo) is described sufficiently to be reproducible
- Whether the number of iterations is sufficient for stable results
- The basis for any country-specific uncertainty ranges that deviate from IPCC defaults

12 Worked example: Country X (hypothetical dairy)

To illustrate the workflow, the following example uses the Country X built-in dataset.

Step 1: Load data: Select "Country X (hypothetical dairy)" from the dropdown on Tab 1. The data loads automatically (12 parameters, dairy smallholder system).

Step 2: QA/QC: Go to Tab 2. All parameters should pass; any Warn flags on `BW` would compare against the continental IPCC default (Annex Tables 10A.1 / 10A.2 / 10A.3), and Warn flags on other parameters reflect non-fatal structural conditions (asymmetric-bounds reminder, sub-category auto-match, etc.), a deliberate country-specific value can be documented in the inventory report rather than corrected.

Step 3: Uncertainty: Go to Tab 3. Note that activity data (N, BW, Milk, WG, Fat) use Normal distributions and coefficients (Ym, DE, Bo, etc.) use PERT or lognormal distributions.

Step 4: Correlations: Go to Tab 4. Select "From template (auto, time-series)", the example dataset includes a 5-year time series from which a Spearman rank correlation matrix is computed automatically (first-difference detrending is applied internally to strip shared long-run growth). The matrix is sampled using Iman–Conover restricted pairing.

Step 5: Simulate: Go to Tab 5. Set iterations to 10,000, GWP to AR6, seed to 42. Click "Run Monte Carlo Simulation".

Step 6: Results: After about 30 seconds, results appear. Note the 95% CI and MoE%. Click "Run Diagnostic" to confirm convergence.

Step 7: Sensitivity: Go to Tab 6. The tornado chart shows which parameters drive most uncertainty. In this example, Ym and BW for cows are typically the dominant contributors.

Step 8: Report: Go to Tab 7. Download the Excel report to see the full IPCC Table 3.3 output.

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